

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE440 Capstone Project

<Project Progress Report>

ProHealth AI Assistant

Intelligent Hospital Portal & Bio_ClinicalBERT Triage System

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1. PROJECT OVERVIEW

1.1. Executive Summary

ProHealth AI Assistant is an end-to-end intelligent hospital management portal and clinical triage referral system developed to eliminate patient misdirection, optimize hospital resource utilization, and accelerate emergency routing. Powered by fine-tuned **Bio_ClinicalBERT** (emilyalsentzer/Bio_ClinicalBERT), the system analyzes unstructured multilingual complaints (Pure Bengali, phonetic Banglish, and English) and maps them across **13 core medical specialties** with confidence probabilities. Simultaneously, a deterministic rule-based urgency triage evaluator identifies critical red-flag triggers across 3 levels (Emergency, Urgent, Routine). The application integrates Google Identity Services OAuth, interactive doctor booking, persistent database tracking, and ReportLab PDF referral tickets, and is fully deployed live on **Render.com**.

1.2. Project Stakeholders

Role / Field	Details
Project Due Date	Final Evaluation / Capstone Submission 2026
Project Core Team (PCT)	Tasmiad Hasan (Lead AI/ML Engineer & Full-Stack Developer)
Other Stakeholders	Course Supervisors, Evaluation Board, Clinical Consultants

1.3. Milestones and Deliverables

ID	Milestone Name	Target Date	Actual Date	Status	Comments
M1	Dataset Collection & Labeling	10 Jul 2026	10 Jul 2026	Achieved	13 specialties annotated
M2	Bio_ClinicalBERT Fine-Tuning	20 Jul 2026	20 Jul 2026	Achieved	77% accuracy / 1.00 F1 key
M3	FastAPI Backend & Triage API	28 Jul 2026	28 Jul 2026	Achieved	Prediction & PDF endpoints
M4	Web Portal & Bilingual I18N	05 Aug 2026	05 Aug 2026	Achieved	Bangla/English toggle & UI
M5	Google Auth & User Dashboard	12 Aug 2026	12 Aug 2026	Achieved	GIS OAuth & persistence
M6	PDF Referral & Cloud Deploy	16 Aug 2026	16 Aug 2026	Achieved	Live on Render.com

1.4. Project Plan (per Work Progress)

Work Package	Planned Start	Planned End	Actual Start	Actual End	Progress (%)
WP1: Data Engineering	01 Jul 2026	10 Jul 2026	01 Jul 2026	10 Jul 2026	100%
WP2: Model Training (Colab)	11 Jul 2026	22 Jul 2026	11 Jul 2026	20 Jul 2026	100%
WP3: Backend REST API	23 Jul 2026	30 Jul 2026	23 Jul 2026	28 Jul 2026	100%
WP4: Web Portal & I18N	31 Jul 2026	08 Aug 2026	31 Jul 2026	05 Aug 2026	100%
WP5: Auth, Dashboard & PDF	09 Aug 2026	15 Aug 2026	09 Aug 2026	14 Aug 2026	100%
WP6: Cloud Deploy & QA	15 Aug 2026	17 Aug 2026	15 Aug 2026	16 Aug 2026	100%

2. PROJECT DETAILS

2.1. Scope Changes

ID	Category	Title & Description	Status	Size	Prio	Approved By	Date
SC 1	Technical	Google OAuth Migration: Switched from Clerk to Google Identity Services (GIS)	Done	3	5	PCT / Sup	15 Aug
SC 2	New Req	Vector PDF Generator: Server-side ReportLab streaming referral tickets	Done	4	4	PCT	14 Aug
SC 3	Technical	Bilingual I18N & Banglish: Dynamic Bangla/English UI & phonetic normalizer	Done	3	5	PCT	08 Aug
SC 4	Technical	Render Cloud Deployment: CPU wheel optimization for free cloud tier	Done	2	4	PCT	16 Aug

Footnotes: Size/Priority scale: 5=Very High, 4=High, 3=Medium, 2=Low, 1=Very Low.

2.2. Major Risks and Actions Taken

ID	Risk Name	Description	Status	L	I	RL	Strategy	Action Details
R1	Cold Start Delay	BERT loading from disk on first web request causing >10s latency	Closed	4	4	16	Reduce	Cached model in RAM at startup
R2	Cloud Size Quota	GPU PyTorch wheels (>2GB) exceeding free cloud memory limits	Closed	5	4	20	Avoid	Used CPU wheel index
R3	Auth Origin Block	Google OAuth blocking requests from unlisted live cloud URLs	Closed	4	5	20	Reduce	Whitelisted Render domain in GCP
R4	Medical Triage Error	Potential misclassification of emergency cases	Closed	2	5	10	Reduce	Added rule-based red-flag layer

Risk Level $RL = L \times I$ (Scale 1-25). Strategy: Avoid, Transfer, Reduce, Accept.

2.3. Major Issues and Actions Taken

ID	Title	Description & Status	Action Taken	Urg	Imp	Size	Owner
I1	Blank PDF Bug	Canvas capture rendering black screen [Resolved]	Switched to ReportLab vector generation	5	5	3	Tasmiad
I2	JSON Persistence	Missing module import skipping booking saves [Resolved]	Imported json and added error handling	5	4	1	Tasmiad
I3	Navbar Stacking	Buttons breaking into 2 rows on laptops [Resolved]	Enforced flexbox nowrap & height:42px	4	3	2	Tasmiad
I4	I18N Desync	Modal text remaining Bangla on English toggle [Resolved]	Populated I18N dict & DOM hooks	4	4	2	Tasmiad

2.4. Other On-Going and Planned Actions

Actions	Due Date	Who & Comments
Interactive Chatbot Triage Assistant: Multi-turn symptom questioning agent	Q4 2026	Tasmiad Hasan — Handling underspecified patient complaints
Hospital EHR / FHIR Protocol Integration: Direct database sync	Q1 2027	Tasmiad Hasan — Integration with hospital management software
Automated SMS & WhatsApp Booking Delivery: Instant SMS notifications	Q1 2027	Tasmiad Hasan — Twilio gateway integration for booking tickets

2.5. Achievements

Project Highlights / Achievements	Comments
State-of-the-Art Clinical Transformer (Bio_ClinicalBERT)	Fine-tuned BERT model on Google Colab, achieving 77.0% overall accuracy and 1.00 F1-score on key departments (Ophthalmology, Urology, Gastroenterology).
End-to-End Hospital Management Web Portal	Built responsive hospital website with 13 doctor directories, facilities explorer, voice recognition, and instant AI triage referral.
Seamless Google Identity OAuth & Dashboard	Implemented 1-tap Google login, appointment persistence in data/appointments.json, and active patient management.
100% Reliable Vector PDF Referral Generator	Implemented server-side ReportLab vector PDF generator embedding patient name, Google ID, department, and doctor recommendations.
Live Production Cloud Deployment	Successfully deployed and hosted live on Render.com (https://prohealth-ai-assistant.onrender.com/) with automated CI/CD from GitHub.